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GENERATIVE AI MINI PROJECT REPORT

On

“Scenario Weaver”

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in

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ABSTRACT

Scenario Weaver is a Generative AI-powered operational simulation and protocol intelligence platform developed using React, Supabase Edge Functions, PostgreSQL, and pgvector. The platform is designed to simulate operational incidents, crisis-response workflows, and protocol-aware decision-making using Retrieval-Augmented Generation (RAG), hybrid semantic retrieval, and AI-driven orchestration.

The system enables users to upload operational documents such as SOPs, emergency response manuals, hazard procedures, and incident reports in PDF or TXT formats. Uploaded documents are processed through a hybrid retrieval pipeline involving semantic vector search, PostgreSQL full-text ranking, and identifier-aware reranking. The platform supports retrieval optimization for operational identifiers including UN hazard codes, NFPA classifications, CAS registry identifiers, and SOP references.

Scenario Weaver integrates AI-generated simulations with event-driven orchestration and persistent agent memory systems to create realistic operational workflows. The platform supports dynamic event progression, chaos engineering, protocol retrieval, simulation evaluation, and automated After-Action Report (AAR) generation.

The project demonstrates the practical application of Generative AI, Retrieval-Augmented Generation, vector databases, and edge-native architectures in operational reasoning and crisis-response modeling.

Keywords: Generative AI, RAG, pgvector, Supabase, Operational Simulation, Hybrid Retrieval, AI Agents, Event-Driven Architecture, Protocol Intelligence

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Chapter 1

INTRODUCTION

1.1 Problem Statement

Modern operational environments such as industrial safety systems, emergency response networks, and crisis-management workflows require rapid and context-aware decision-making. Traditional simulation systems rely heavily on static workflows and predefined rule engines that lack contextual reasoning and adaptability.

Organizations often struggle with:

- Retrieving relevant protocols during operational incidents
- Simulating dynamic crisis-response workflows
- Evaluating operational decisions effectively
- Managing event-driven escalation chains
- Integrating AI reasoning into lightweight deployment environments

Scenario Weaver addresses these limitations by combining Generative AI, Retrieval-Augmented Generation (RAG), hybrid semantic retrieval, and event-driven orchestration into a unified operational simulation platform.

1.2 Objectives

- To develop a Generative AI-powered operational simulation platform.
- To implement Retrieval-Augmented Generation using pgvector.
- To support protocol-aware operational reasoning.
- To build hybrid semantic and keyword retrieval pipelines.
- To simulate dynamic operational events and escalation chains.
- To generate automated evaluations and After-Action Reports.
- To design an edge-native scalable architecture.

1.3 Scope

The scope of Scenario Weaver includes:

- Crisis-response simulations
- Operational protocol retrieval
- Hazard management workflows
- AI-assisted operational decision-making
- Simulation evaluation and reporting
- Agent memory and event orchestration

1.4 Methodology

The project follows a modular architecture combining frontend development, edge-native backend orchestration, vector retrieval systems, and Generative AI workflows.

The methodology includes:

1. Protocol ingestion and parsing
2. Text chunk generation
3. Embedding generation
4. Vector indexing using pgvector
5. Hybrid retrieval implementation
6. Simulation event orchestration
7. Evaluation and reporting

Chapter 2

LITERATURE SURVEY

2.1 Existing Systems

Existing operational simulation systems primarily rely on static workflows and predefined rule engines. Traditional crisis-management systems provide limited contextual reasoning and lack intelligent retrieval capabilities.

Most conventional systems suffer from:

- Limited semantic understanding
- No Retrieval-Augmented Generation support
- Poor protocol retrieval accuracy
- Lack of operational memory systems
- Limited simulation adaptability

2.2 Proposed Solution

Scenario Weaver introduces a lightweight edge-native Generative AI platform integrating:

- Retrieval-Augmented Generation (RAG)
- pgvector-based semantic search
- Hybrid retrieval pipelines
- Event-driven simulation workflows
- Agent memory systems
- Automated evaluation and AAR generation

The proposed system improves contextual reasoning, operational retrieval accuracy, and simulation realism.

Chapter 3

REQUIREMENT ANALYSIS AND SPECIFICATION

3.1 Functional Requirements

Sl.No	Requirement Description
1	The system must allow users to upload PDF and TXT operational documents.
2	The system must perform semantic retrieval using pgvector.
3	The system must support hybrid retrieval pipelines.
4	The system must simulate operational events dynamically.
5	The system must maintain agent memory and event history.
6	The system must generate evaluation reports and AARs.
7	The system must support retrieval debugging and protocol inspection.
8	The system must support scalable edge-native deployment.

3.2 Software Requirements

- React
- TypeScript
- Tailwind CSS
- Supabase
- PostgreSQL
- pgvector
- Deno Runtime
- Node.js
- VS Code

Chapter 4

SYSTEM DESIGN

4.1 System Architecture

Scenario Weaver follows an edge-native Generative AI architecture designed for operational simulations, protocol intelligence, and AI-driven forecasting workflows. The platform integrates Retrieval-Augmented Generation (RAG), hybrid semantic retrieval, event-driven orchestration, and relationship graph modeling into a unified simulation environment.

The system enables users to:

- Create AI-powered operational simulations
- Upload and retrieve operational protocols
- Generate relationship graphs between entities
- Simulate multi-agent interactions
- Perform event-driven forecasting
- Generate evaluations and After-Action Reports

4.2 Purpose of the System

The primary purpose of Scenario Weaver is to provide an intelligent operational reasoning platform capable of simulating crisis-response workflows, forecasting operational outcomes, and analyzing relationships between agents, systems, and hazards.

The system combines Generative AI and hybrid retrieval pipelines to improve contextual reasoning and decision-support capabilities.

4.3 System Modules

4.3.1 Simulation Engine

The simulation engine dynamically generates operational scenarios using Generative AI models. It supports:

- Multi-agent simulations
- Event progression
- Scenario branching
- Forecast generation
- Escalation modeling

4.3.2 Protocol Intelligence Module

This module allows users to upload operational documents such as SOPs, incident manuals, hazard guidelines, and emergency-response protocols.

The system performs:

- Document parsing
- Text chunking
- Embedding generation
- Semantic indexing
- Hybrid retrieval

4.3.3 Relationship Graph Module

The relationship graph module models dependencies and interactions between operational entities, institutions, actors, hazards, and events.

The graph supports:

- Influence analysis
- Dependency tracking
- Contagion modeling
- Escalation analysis
- Causal reasoning

4.3.4 Evaluation Module

The evaluation engine analyzes generated simulations and produces:

- Operational assessments
- Decision evaluations
- Failure summaries
- Timeline reconstruction
- After-Action Reports (AAR)

4.4 Workflow of the System

1. User enters a custom operational scenario.
2. The system performs input analysis and actor identification.
3. Relevant protocols and operational documents are retrieved.
4. AI agents generate operational forecasts and event chains.
5. Relationship graphs are generated dynamically.

6. Simulations evolve using event-driven orchestration.
7. Evaluations and reports are generated automatically.

4.5 Advantages of the System

- Supports real-time operational reasoning
- Improves protocol retrieval accuracy
- Enables AI-powered crisis simulations
- Provides relationship graph visualization
- Supports scalable edge-native deployment
- Reduces hallucinations using RAG grounding

4.6 System Architecture Layers

4.6.1 Presentation Layer

Built using:

- React
- Vite
- TypeScript
- Tailwind CSS

This layer handles:

- Simulation interfaces
- Protocol uploads
- Graph visualization
- Dashboard analytics
- Retrieval debugging

4.6.2 Application Layer

The application layer is implemented using Supabase Edge Functions and Deno runtime.

It performs:

- Hybrid retrieval
- Simulation orchestration
- Event management
- Relationship graph generation
- Evaluation processing

Hybrid Retrieval Formula:

$$Final\ Score = (0.7 \times Vector\ Similarity) + (0.3 \times Keyword\ Rank) + Identifier\ Boost$$

4.6.3 Data Layer

The data layer uses PostgreSQL and pgvector for:

- Vector storage
- Semantic retrieval
- Event persistence
- Agent memory
- Simulation history

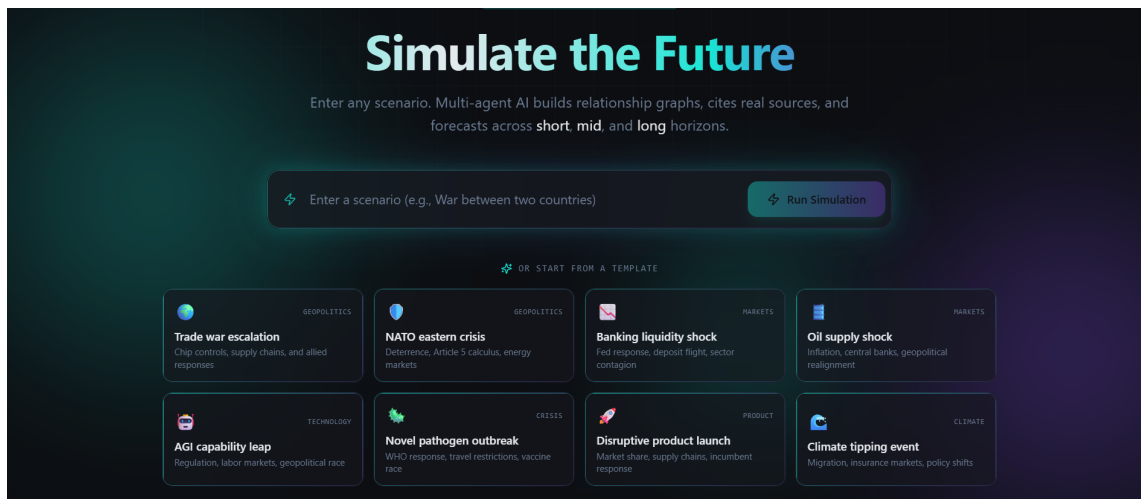


Figure 4.1: Scenario Weaver Landing Page

The landing page provides an interactive interface for creating AI-driven operational simulations. Users can enter custom scenarios or select predefined templates such as banking crises, geopolitical conflicts, climate events, and supply-chain disruptions. The interface demonstrates the platform’s ability to support multi-domain operational forecasting and Generative AI-based scenario modeling.

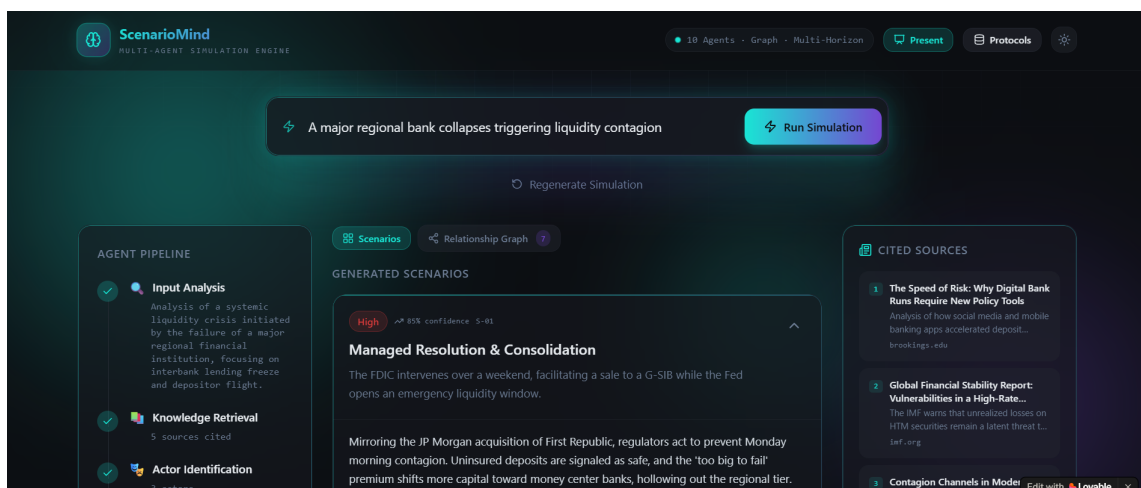


Figure 4.2: Simulation Dashboard and Agent Pipeline

The simulation dashboard displays AI-generated operational scenarios, agent pipelines, retrieval workflows, and cited knowledge sources. The system performs input analysis, protocol retrieval, actor identification, and event orchestration to generate context-aware operational forecasts. The dashboard also supports multi-horizon simulation reasoning and source-grounded AI outputs.



Figure 4.3: Relationship Graph Visualization

The relationship graph visualizes dependencies and interactions between operational entities, institutions, actors, and hazards. The graph engine models influence chains, dependencies, neutral relationships, and escalation pathways across multiple agents and operational systems. This enables causal reasoning, contagion analysis, and event propagation tracking during simulations.

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Implementation

Scenario Weaver is implemented as a Generative AI-powered web platform.

The frontend uses React and TypeScript for responsive UI development. Supabase Edge Functions provide lightweight backend orchestration. PostgreSQL and pgvector are used for semantic vector retrieval.

The document ingestion pipeline performs:

1. PDF/TXT parsing
2. Text extraction
3. Chunk generation
4. Embedding creation
5. Vector indexing
6. Metadata storage

The simulation engine generates AI-driven operational workflows using event-based orchestration.

5.2 Testing

5.2.1 Functional Testing

The following modules were tested:

- Protocol uploads
- Hybrid retrieval
- Simulation execution
- Event generation
- Evaluation pipelines
- AAR generation

5.2.2 Performance Testing

Performance testing was conducted for:

- Retrieval latency
- Vector search performance
- Simulation response time
- Dashboard rendering

5.3 Results

The platform successfully:

- Retrieves operational protocols accurately
- Generates AI-driven simulations
- Maintains event memory
- Supports hazard-code reranking
- Produces automated evaluations and AARs

Chapter 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

Scenario Weaver successfully demonstrates the integration of Generative AI, Retrieval-Augmented Generation, hybrid retrieval systems, and event-driven orchestration into a unified operational simulation platform.

The project provides protocol-aware operational reasoning, scalable retrieval systems, AI-generated simulations, and automated evaluation pipelines while maintaining lightweight edge-native deployment architecture.

The platform highlights the practical application of Generative AI in operational intelligence, crisis-response modeling, and simulation-based decision support systems.

6.2 Future Enhancements

Future improvements include:

- Real-time simulation streaming
- Multi-agent orchestration
- Knowledge graph visualization
- Temporal event analytics
- Voice-assisted simulations
- Advanced reranking models
- Multimodal document understanding